

A CONDITIONAL REDUCTION OF A CHARACTERISTIC-FIVE COMMON-COVER OBSTRUCTION

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ABSTRACT. Let $k = \overline{\mathbb{F}}_5$ and $S = \mathbb{P}_k^1(31, 31, 31)$. The common-cover counterexample considered here is reduced to the visibility of finite étale self-correspondences of S . The reduction is then carried to the curve level: a non-visible correspondence is assumed to yield a particular genus-11, degree-35 pair of three-point maps, and the remaining work is to exclude three explicit branches of this pair. One branch is further reduced to two stated computer-algebra claims. These four open theorems together imply the visibility theorem for S , hence the desired common-cover obstruction.

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1. STATEMENT OF THE REDUCTION

The original question is the following.

Question 1.1 (Common-cover question over $\overline{\mathbb{F}}_5$). *Do any two smooth proper connected curves of genus at least 2 over $k = \overline{\mathbb{F}}_5$ admit a common connected finite étale cover?*

The proposed counterexample is $Y : y^{31} = x(x - 1)$. Put $S = \mathbb{P}_k^1(31, 31, 31)$. Section 2 proves that it is enough to show every connected finite étale self-correspondence of S is visible, in the sense of Definition 2.3. The curve-level part of the note reduces this visibility theorem to Open Theorems 3.5, 6.1, 7.1, and 8.5; the formal implication is Theorem 9.3.

2. THE GLOBAL STACK ROUTE

Let $k = \overline{\mathbb{F}}_5$. The stack

$$S = \mathbb{P}_k^1(31, 31, 31)$$

is the smooth proper stacky curve with coarse space $\mathbb{P}_k^1$ and stabilizer order 31 at $0, 1, \infty$.

Lemma 2.1 (The cyclic atlas). *Let*

$$Y : y^{31} = x(x-1)$$

and let C_{31} act by $y \mapsto \zeta y$. Then

$$S = [Y/C_{31}],$$

the map $Y \rightarrow S$ is a representable finite etale C_{31} -torsor, and $g(Y) = 15$.

Proof. The coarse map $Y \rightarrow \mathbb{P}_x^1$ has degree 31 and is totally ramified over $0, 1, \infty$. Riemann–Hurwitz gives

$$2g(Y) - 2 = 31(-2) + 3(31 - 1) = 28,$$

so $g(Y) = 15$. The quotient stack $[Y/C_{31}]$ remembers the three coarse ramification stabilizers; therefore the quotient map $Y \rightarrow [Y/C_{31}]$ is finite etale in the stack sense and representable. $\square$

The three stacky points have the same order, so S_3 acts on S by permuting them. The quotient is

$$S_0 = S/S_3 \simeq \mathbb{P}_k^1(2, 3, 62),$$

and the quotient map is denoted

$$\pi_0 : S \longrightarrow S_0.$$

It is finite etale of degree 6 as a morphism of stacky curves: the branch orders of the coarse S_3 -quotient are exactly absorbed by the stack indices $2, 3, 62$ on the target.

Definition 2.2 (Finite etale self-correspondence). *A finite etale self-correspondence of S is a connected smooth proper Deligne–Mumford curve T together with two representable finite etale maps*

$$u, v : T \rightarrow S.$$

Definition 2.3 (Visible self-correspondence). *A finite etale self-correspondence $u, v : T \rightarrow S$ is visible if*

$$\pi_0 \circ u \simeq \pi_0 \circ v.$$

Equivalently, $(u, v) : T \rightarrow S \times S$ factors through the fiber product

$$S \times_{S_0} S.$$

Since $S \rightarrow S_0$ is the S_3 -quotient and T is connected, this is the same as saying that $v = \sigma \circ u$ for one fixed $\sigma \in S_3$.

The assertion that all finite etale self-correspondences of S are visible is the visibility theorem for S . In the usual shorthand it is written

$$\mathrm{Comm}_{\mathrm{alg}, k}(S) = \mathrm{Aut}(S) = S_3.$$

Lemma 2.4 (No genus-three etale cover of S_0). *No smooth proper connected genus-three curve admits a finite etale map to*

$$S_0 = \mathbb{P}_k^1(2, 3, 62).$$

Proof. For a stacky projective line $\mathbb{P}^1(a, b, c)$,

$$\deg K_{\mathbb{P}^1(a, b, c)} = -2 + \left(1 - \frac{1}{a}\right) + \left(1 - \frac{1}{b}\right) + \left(1 - \frac{1}{c}\right).$$

For $(a, b, c) = (2, 3, 62)$ this is $14/93$. If $X \rightarrow S_0$ were finite etale of degree d and $g(X) = 3$, then

$$4 = \deg K_X = d \cdot \frac{14}{93},$$

so $d = 186/7$, impossible. $\square$

Proposition 2.5 (Conditional common-cover counterexample). *Assume every finite etale self-correspondence of S is visible. Then $Y : y^{31} = x(x-1)$ has no connected common finite etale cover with any smooth proper connected genus-three curve over k .*

Proof. Suppose, to the contrary, that $Z \rightarrow X$ and $Z \rightarrow Y$ are connected finite etale covers with $g(X) = 3$. Compose $Z \rightarrow Y$ with the atlas torsor $Y \rightarrow S$, and call the resulting finite etale map $b : Z \rightarrow S$.

Let

$$R = Z \times_X Z$$

with projections $p_1, p_2 : R \rightrightarrows Z$. For each connected component R_α of R , the two maps

$$b \circ p_1, b \circ p_2 : R_\alpha \rightarrow S$$

form a finite etale self-correspondence of S . By Definition 2.3,

$$\pi_0 \circ b \circ p_1 = \pi_0 \circ b \circ p_2$$

on every R_α , hence on all of R . Thus $\pi_0 \circ b : Z \rightarrow S_0$ is invariant under the finite etale equivalence relation $R \rightrightarrows Z$. Since $Z \rightarrow X$ is finite etale surjective, effective descent for finite etale morphisms gives a finite etale map

$$X \rightarrow S_0.$$

This contradicts Lemma 2.4. $\square$

Remark 2.6. *The proposition reduces the common-cover counterexample to the algebraic visibility theorem for S . The rest of the note reduces that theorem to the four open theorems assembled in Theorem 9.3.*

3. PROFILE PAIRS

Definition 3.1 (Three-point profile map). *Let $n, m \geq 1$. A separable function $h : C \rightarrow \mathbb{P}^1$ on a smooth proper connected curve has three-point profile $(n; m)$ if $\deg h = n + m$, all ramification of h lies over $0, 1, \infty$, and for each $a \in \{0, 1, \infty\}$ one has*

$$h^{-1}(a) = nP_a(h) + E_a(h),$$

where $P_a(h)$ is one point and $E_a(h)$ is a reduced effective divisor of degree m , disjoint from the two other E -divisors for the same map. In the rest of the note, “profile map” means three-point profile $(31; 4)$.

Lemma 3.2 (Genus of a $(31; 4)$ -profile map). *If C admits a $(31; 4)$ -profile map, then $g(C) = 11$.*

Proof. The ramification contribution over each of $0, 1, \infty$ is $31 - 1 = 30$. Since there is no other ramification,

$$2g(C) - 2 = 35(-2) + 3 \cdot 30 = 20,$$

so $g(C) = 11$. $\square$

Definition 3.3 (Profile-4 two-map pair). *A profile-4 two-map pair is a smooth proper connected curve C/k with two $(31; 4)$ -profile maps $x, r : C \rightarrow \mathbb{P}^1$, such that*

$$k(C) = k(x, r),$$

the twelve unramified boundary points agree as a reduced divisor

$$U = E_0 + E_1 + E_\infty = F_0 + F_1 + F_\infty,$$

where $E_i = E_i(x)$ and $F_j = E_j(r)$, and the incidence matrix

$$M_{ij} := \deg(E_i \cap F_j)$$

is

$$M = \begin{pmatrix} 0 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & 0 \end{pmatrix} \quad (i, j \in \{0, 1, \infty\}).$$

Lemma 3.4 (Divisors attached to a profile-4 pair). *Let (C, x, r) be a profile-4 pair. With*

$$P_i = P_i(x), \quad Q_j = P_j(r),$$

one has

$$\operatorname{div}(x) = 31P_0 + E_0 - 31P_\infty - E_\infty.$$

Moreover

$$\operatorname{div}(x - 1) = 31P_1 + E_1 - 31P_\infty - E_\infty,$$

and

$$\operatorname{div}(r) = 31Q_0 + F_0 - 31Q_\infty - F_\infty, \quad \operatorname{div}(r - 1) = 31Q_1 + F_1 - 31Q_\infty - F_\infty.$$

Proof. For x , the zero divisor is the scheme-theoretic fiber over 0, namely

$$x^{-1}(0) = 31P_0 + E_0,$$

and the pole divisor is the fiber over ∞ ,

$$x^{-1}(\infty) = 31P_\infty + E_\infty.$$

Their difference is $\operatorname{div}(x)$. The same argument applied to $x - 1$ uses the fiber of x over 1 for the zeros and the same pole fiber as x . The two formulas for r are identical with P, E, x replaced by Q, F, r . $\square$

Write the nonempty atoms of the incidence decomposition as

$$\begin{aligned} A &= U_{0,1}, & B &= U_{0,\infty}, & C &= U_{1,0}, \\ D &= U_{1,1}, & E &= U_{1,\infty}, & F &= U_{\infty,0}, \\ G &= U_{\infty,1}. \end{aligned}$$

Their degrees are

$$\deg A = \deg C = \deg E = \deg G = 1, \quad \deg D = 2, \quad \deg B = \deg F = 3.$$

The atoms $U_{0,0}$ and $U_{\infty,\infty}$ are empty.

Open theorem 3.5 (A. Profile extraction from a non-visible correspondence). *Let*

$$u, v : T \rightarrow S = \mathbb{P}_k^1(31, 31, 31)$$

be a connected representable finite etale self-correspondence such that

$$\pi_0 \circ u \not\sim \pi_0 \circ v.$$

Then, after possibly interchanging u and v and composing one of them with a visible S_3 -symmetry of S , there is a smooth proper connected curve C/k and separable functions $x, r \in k(C)$ such that (C, x, r) is a profile-4 two-map pair in the sense of Definition 3.3.

Remark 3.6. *Open Theorem 3.5 is the global-to-curve reduction: it replaces a non-visible stack correspondence by the explicit two-map incidence datum of Definition 3.3.*

Remark 3.7 (Why profile 4). *Among 3×3 nonnegative integer matrices with all row and column sums 4, modulo independent relabeling of rows and columns, there are nine incidence profiles. The profile above is the smallest of the six profiles in which no r -boundary block is equal to E_∞ . This matters for the Kummer tests in Section 4. The unproved content in Open Theorem 3.5 is that, after the visible S_3 -symmetries and the preceding reductions, no other incidence profile has to be considered.*

4. LOGARITHMIC DIFFERENTIALS AND KUMMER TESTS

We use the following logarithmic identities.

4.1. Logarithmic forms. Let h be one profile map. For $c \in \{0, 1, \infty\}$, define

$$\Omega_c(h) = \frac{dh}{(h-a)(h-b)}, \quad \{a, b, c\} = \{0, 1, \infty\}.$$

Thus

$$\Omega_0(h) = \frac{dh}{h-1}, \quad \Omega_1(h) = \frac{dh}{h}, \quad \Omega_\infty(h) = \frac{dh}{h(h-1)}.$$

Lemma 4.1 (Divisor of Ω_c). *For $\{a, b, c\} = \{0, 1, \infty\}$,*

$$\operatorname{div} \Omega_c(h) = 30P_c(h) - P_a(h) - P_b(h) - E_a(h) - E_b(h).$$

Proof. At $P_c(h)$, use $h-c$ as a target coordinate if $c \neq \infty$, and use $1/h$ if $c = \infty$. In either coordinate the local form of the map is $t \mapsto t^{31}$, so dh contributes order 30 relative to a local parameter on C , while neither denominator factor vanishes. Hence $\Omega_c(h)$ has order 30 at $P_c(h)$.

At $P_a(h)$, the corresponding denominator factor has order 31, while dh has order 30, so the order is -1 ; the same argument applies at $P_b(h)$. At each point of $E_a(h)$ or $E_b(h)$, the map is unramified and the corresponding denominator factor has a simple zero, so the order is -1 . Away from the three boundary fibers the denominator is nonzero and the map is unramified, so there are no further zeros or poles. The displayed divisor also has the required degree

$$30 - 1 - 1 - 4 - 4 = 20 = 2g(C) - 2.$$

□

At an unramified boundary point with boundary label ℓ , the residue of $\Omega_c(h)$ is

	$\ell = 0$	$\ell = 1$	$\ell = \infty$
$c = 0$	0	1	-1
$c = 1$	1	0	-1
$c = \infty$	-1	1	0.

For example, near a point with $h = 0$, the form dh/h has residue 1, while $dh/(h-1)$ is regular.

Multiplying the three logarithmic forms for one map gives

$$\operatorname{div}(\Omega_0(h)\Omega_1(h)\Omega_\infty(h)) = 28(P_0(h) + P_1(h) + P_\infty(h)) - 2U_h,$$

where $U_h = E_0(h) + E_1(h) + E_\infty(h)$. Hence a profile-4 pair satisfies the Picard constraint

$$28((P_0 + P_1 + P_\infty) - (Q_0 + Q_1 + Q_\infty)) = 0 \quad \text{in } \operatorname{Pic}^0(C).$$

4.2. Kummer valuation tests. Let $F = k(C)$ and let

$$K = F(y), \quad y^{31} = x(x-1).$$

Lemma 4.2 (Boundary valuation obstruction). *None of*

$$r(r-1), \quad \frac{1-r}{r^2}, \quad \frac{r}{(r-1)^2}$$

is a 31st power in K .

Proof. Because k is algebraically closed and contains μ_{31} , Kummer theory says that an element $B \in F^\times$ becomes a 31st power in K only if its class in $F^\times/F^{\times 31}$ lies in the subgroup generated by $x(x-1)$. Equivalently,

$$B = (x(x-1))^m a^{31}$$

for some $m \in \mathbb{Z}/31\mathbb{Z}$ and $a \in F^\times$. Therefore the valuation vector of B on the twelve points of U , modulo 31, must be m times the valuation vector of $x(x-1)$, namely

$$1 \text{ on } E_0, \quad 1 \text{ on } E_1, \quad -2 \text{ on } E_\infty.$$

The three displayed functions have valuation vectors

	F_0	F_1	F_∞
$r(r-1)$	1	1	-2
$(1-r)/r^2$	-2	1	1
$r/(r-1)^2$	1	-2	1.

If one of these vectors equals $m(1, 1, -2)$ after pulling back to the E -partition of U , then the unique four-point block on which the value is -2 must be E_∞ . The alternative $m \neq 1$ is impossible: then the two values m and $-2m$ would have to be among $\{1, -2\}$, and the only such scalar preserving the multiplicities 8 and 4 is $m = 1$.

Thus the three necessary conditions are respectively

$$F_\infty = E_\infty, \quad F_0 = E_\infty, \quad F_1 = E_\infty.$$

For profile 4, no column of M is $(0, 0, 4)^t$, so none of F_0, F_1, F_∞ equals E_∞ . All three elements therefore fail the valuation test. $\square$

4.3. A characteristic-five p -basis constraint. Let $F = k(C)$. Since x is separable and k is perfect of characteristic 5,

$$F = F^5(x).$$

Thus

$$r = u_0^5 + u_1^5 x + u_2^5 x^2 + u_3^5 x^3 + u_4^5 x^4$$

for unique $u_i \in F$, and

$$\frac{dr}{dx} = u_1^5 + 2u_2^5 x + 3u_3^5 x^2 + 4u_4^5 x^3.$$

At an unramified x -pole $P \in E_\infty$, the valuations of the five terms $u_i^5 x^i$ have distinct residues modulo 5. Therefore there is no cancellation among them.

As a check on possible characteristic-5 constructions, suppose

$$r = x + u^5$$

and suppose r is again a profile map with branch values $0, 1, \infty$. Any pole of u away from $x^{-1}(\infty)$ would give r a pole whose order is divisible by 5, which is incompatible with the profile. At the four simple poles in E_∞ , the same residue-modulo-5 argument shows that u is regular. At P_∞ , the

pole order of u is at most 6, otherwise u^5 would dominate the order-31 pole of x . Thus $\deg u \leq 6$. At a finite high point of x , say $x = a + t^{31}$, the equality

$$r - r(P) = t^{31} + (u - u(P))^5$$

has tame ramification index 31 only if $\text{ord}_P(u - u(P)) \geq 7$, unless u is constant. This contradicts $\deg u \leq 6$. Hence u is constant, and then the branch-value condition forces that constant to be 0.

5. HIGH-POINT COINCIDENCES

A high-point coincidence is an equality $Q_j = P_i$ among the six ramified points of the two profile maps. Its entry is M_{ij} ; for instance an entry-one coincidence means $M_{ij} = 1$.

Lemma 5.1 (High-point quotient). *If $Q_j = P_i$, then*

$$Y_{i,j} := \frac{\Omega_j(r)}{\Omega_i(x)}$$

has degree at most $6 - M_{ij}$.

Proof. Let $\{a, b, i\} = \{0, 1, \infty\}$ and $\{c, d, j\} = \{0, 1, \infty\}$. By Lemma 4.1, the order 30 terms at the common point $Q_j = P_i$ cancel, and

$$\text{div}(Y_{i,j}) = (P_a + P_b + F_j) - (Q_c + Q_d + E_i).$$

The positive part has degree at most

$$2 + \deg(F_j - E_i)_+ = 2 + 4 - M_{ij} = 6 - M_{ij}.$$

Any additional high-point coincidence only cancels more terms. □

At every atom $U_{\alpha,\beta}$ with $\alpha \neq i$ and $\beta \neq j$, both logarithmic forms have simple poles, so the value of $Y_{i,j}$ is the quotient of the two residues in the residue table. Counting these reduced atoms gives the following forced finite fibers:

coincidence	M_{ij}	$\deg Y_{i,j} \leq$	forced finite values
$Q_0 = P_0$	0	6	$1 : 2, -1 : 2$
$Q_1 = P_0$	1	5	$1 : 1, -1 : 4$
$Q_\infty = P_0$	3	3	$1 : 5, -1 : 2$
$Q_0 = P_1$	1	5	$1 : 1, -1 : 4$
$Q_1 = P_1$	2	4	$-1 : 6$
$Q_\infty = P_1$	1	5	$1 : 4, -1 : 1$
$Q_0 = P_\infty$	3	3	$1 : 5, -1 : 2$
$Q_1 = P_\infty$	1	5	$1 : 4, -1 : 1$
$Q_\infty = P_\infty$	0	6	$1 : 2, -1 : 2$

Hence the coincidences

$$Q_\infty = P_0, \quad Q_1 = P_1, \quad Q_0 = P_\infty$$

are impossible.

Lemma 5.2 (Simultaneous inversion). *The case $Q_\infty = P_\infty$ is equivalent to the case $Q_0 = P_0$.*

Proof. Replace x, r by

$$x' = 1/x, \quad r' = 1/r.$$

Then

$$\text{div}(x') = 31P_\infty + E_\infty - 31P_0 - E_0,$$

and

$$\text{div}(x' - 1) = \text{div}((1 - x)/x) = 31P_1 + E_1 - 31P_0 - E_0.$$

Thus the labels 0 and ∞ are interchanged for x , while the label 1 is fixed. The same calculation applies to r . The incidence matrix is unchanged by simultaneous interchange of rows $0, \infty$ and columns $0, \infty$:

$$\begin{pmatrix} 0 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & 0 \end{pmatrix}.$$

Therefore $Q_\infty = P_\infty$ for (x, r) becomes $Q'_0 = P'_0$ for (x', r') , and the operation is an involution. $\square$

Lemma 5.3 (The paired entry-zero edge is impossible). *There is no profile-4 pair with*

$$Q_0 = P_0, \quad Q_\infty = P_\infty.$$

Proof. Assume both equalities. Put

$$u = \frac{\Omega_0(r)}{\Omega_0(x)}, \quad v = \frac{\Omega_\infty(r)}{\Omega_\infty(x)}.$$

The divisor formula gives

$$\operatorname{div}(u) = P_1 + U_{1,0} + U_{\infty,0} - Q_1 - U_{0,1} - U_{0,\infty},$$

and

$$\operatorname{div}(v) = P_1 + U_{0,\infty} + U_{1,\infty} - Q_1 - U_{\infty,0} - U_{\infty,1}.$$

Thus both u and v have degree 5.

Let $\Gamma \subset \mathbb{P}_u^1 \times \mathbb{P}_v^1$ be the image of (u, v) , and let $n = [k(C) : k(u, v)]$. Since both functions have degree 5, the bidegree of Γ is $(5/n, 5/n)$, so $n = 1$ or 5. If $n = 5$, then Γ has bidegree $(1, 1)$, hence $v = M(u)$ for a linear fractional transformation. The points P_1, Q_1 , and the two points of $U_{1,1}$, force

$$M(0) = 0, \quad M(\infty) = \infty, \quad M(1) = 1,$$

so M is the identity. But $U_{0,\infty}$ maps to $(\infty, 0)$, contradiction. Therefore $n = 1$, and C is the normalization of a bidegree- $(5, 5)$ curve Γ .

The atoms $U_{0,\infty}$, $U_{\infty,0}$, and $U_{1,1}$ map to

$$(\infty, 0), \quad (0, \infty), \quad (1, 1),$$

with branch counts 3, 3, 2: the corresponding atoms are reduced on the normalization C , and $C \rightarrow \Gamma$ is birational. A singularity of a reduced curve on a smooth surface with m distinct analytic branches has $\delta \geq \binom{m}{2}$. Hence these three points contribute at least

$$\binom{3}{2} + \binom{3}{2} + \binom{2}{2} = 7$$

to the total δ -invariant. The arithmetic genus of a bidegree- $(5, 5)$ curve in $\mathbb{P}^1 \times \mathbb{P}^1$ is 16, so

$$g(C) \leq 16 - 7 = 9,$$

contradicting Lemma 3.2. $\square$

Corollary 5.4 (Entry-zero coverage modulo entry-one). *Assume $Q_0 = P_0$. If no entry-one high-point coincidence occurs, then no other high-point coincidence occurs.*

Proof. The high-point quotient table already excludes

$$Q_\infty = P_0, \quad Q_1 = P_1, \quad Q_0 = P_\infty.$$

The equalities $Q_1 = P_0$ and $Q_0 = P_1$ would identify distinct high points of a single map. The remaining possible extra coincidences are

$$Q_1 = P_\infty, \quad Q_\infty = P_1, \quad Q_\infty = P_\infty.$$

The first two are entry-one coincidences, and the last is impossible by Lemma 5.3. $\square$

6. THE ENTRY-ONE BRANCH

The four entry-one coincidences are equivalent under the symmetries of the incidence matrix and by interchanging the two maps. We use the representative

$$Q_1 = P_0.$$

6.1. The degree-five quotient. Define

$$z = \frac{\Omega_1(r)}{\Omega_0(x)} = \frac{dr/r}{dx/(x-1)}.$$

By Lemma 4.1,

$$\operatorname{div}(z) = P_1 + P_\infty + D + G - Q_0 - Q_\infty - B.$$

Thus $\deg z = 5$ in the no-further-highpoint subcase. If another high-point coincidence is also present, the divisor above has further cancellation; such lower-degree specializations remain part of the entry-one problem. We first treat the no-further-highpoint case.

The boundary residues force

$$z = 1 \text{ on } C, \quad z = -1 \text{ on } E + F.$$

Let

$$\alpha = z(P_0) = z(Q_1), \quad u = z(A), \quad c = x(Q_0), \quad d = x(Q_\infty).$$

The points Q_0 and Q_∞ may lie in the same x -fiber; no condition $c \neq d$ is imposed.

Since $\deg x = 35$ and $\deg z = 5$, if $k(x, z) \neq k(C)$ then the degree $[k(C) : k(x, z)]$ divides both 35 and 5. The nontrivial possibility would force $x \in k(z)$, so the ramification index 31 of x would factor through a degree-5 map, impossible. Hence

$$k(C) = k(x, z),$$

and C is the normalization of an integral curve of bidegree $(5, 35)$ in $\mathbb{P}_x^1 \times \mathbb{P}_z^1$.

6.2. Norms and the tautological trace identity. Write $\operatorname{Norm}_z = \operatorname{Norm}_{k(C)/k(z)}$. Pushing the divisors of $x, x-1, r, r-1$ along z gives, up to constants,

$$\operatorname{Norm}_z(x) = \frac{(z-\alpha)^{31}(z-u)}{z^{32}(z+1)^3}, \quad \operatorname{Norm}_z(x-1) = \frac{z(z-1)}{(z+1)^2},$$

and

$$\operatorname{Norm}_z(r) = (z-1)(z+1)^2, \quad \operatorname{Norm}_z(r-1) = \frac{(z-\alpha)^{31}(z-u)z^3}{z+1}.$$

The defining differential relation is

$$d \log r = z d \log(x-1).$$

Taking trace to $k(z)$ gives

$$d \log \operatorname{Norm}_z(r) = z d \log \operatorname{Norm}_z(x-1).$$

Substitution yields

$$\frac{1}{z-1} + \frac{2}{z+1} = z \left(\frac{1}{z} + \frac{1}{z-1} - \frac{2}{z+1} \right),$$

which is an identity. Thus the first pushed-forward norm/trace relation is not an obstruction.

6.3. The p -closed equation. Let

$$\partial = (x-1) \frac{d}{dx}.$$

For this entry-one quotient, the Cartier-operator formula for the logarithmic exact differential $d \log r = z d \log(x-1)$ gives the characteristic-5 equation

$$\partial^4 z = z - z^5.$$

This at least proves separability of z . If $dz = 0$, then $\partial z = 0$, so the equation gives $z^5 = z$. Over $k = \mathbb{F}_5$, this forces $z \in \mathbb{F}_5$, contradicting $\deg z = 5$.

6.4. A bidegree-(5, 35) normal form. Normalize the bidegree equation $f(x, z) = 0$ by

$$[z^{35}]f = x^3(x-c)(x-d), \quad [x^5]f = z^{32}(z+1)^3.$$

Put

$$s = c + d, \quad p = cd, \quad N = (1-c)(1-d).$$

Here $N \neq 0$; otherwise the fiber $x = 1$ would contain a pole of z , contradicting the displayed divisor of z . Let

$$\Phi(z) = (z-\alpha)^{31}(z-u), \quad \Psi(z) = z^{33}(z-1)(z+1), \quad \Gamma(z) = z^{32}(z+1)^3.$$

Then one may write

$$f(x, z) = (1-x)K\Phi(z) + xN\Psi(z) + x(x-1)H(x, z),$$

where

$$H = h_0(z) + xh_1(z) + x^2h_2(z) + x^3\Gamma(z).$$

The leading z^{35} -coefficients are forced:

$$[z^{35}]h_0 = N, \quad [z^{35}]h_1 = N, \quad [z^{35}]h_2 = 1 - s.$$

Indeed the coefficient of z^{35} is

$$xN + x(x-1)(N + Nx + (1-s)x^2 + x^3) = x^3(x^2 - sx + p).$$

The fiber $x = 0$ gives

$$f(0, z) = K(z-\alpha)^{31}(z-u),$$

the fiber $x = 1$ gives

$$f(1, z) = Nz^{33}(z-1)(z+1),$$

and the $x = \infty$ boundary is encoded by $[x^5]f = \Gamma(z)$.

At the corner $(x, z) = (0, \infty)$, write $w = 1/z$ and

$$F(x, w) = w^{35}f(x, 1/w).$$

The three points of $B = U_{0,\infty}$ have local behavior $w \sim x$. Thus the tangent cubic is proportional to

$$(w-x)^3.$$

Writing

$$w^{35}h_0(1/w) = N + aw + bw^2 + O(w^3), \quad w^{35}h_1(1/w) = N + ew + O(w^2),$$

the degree-two term forces $a = 0$, and the cubic comparison gives

$$p = -K, \quad e = -3K, \quad b = 3K - N.$$

These are genuine constraints, but they are not yet contradictory.

Open theorem 6.1 (B. Entry-one exclusion). *No profile-4 pair has an entry-one high-point coincidence. In the representative $Q_1 = P_0$ with no further high-point coincidence, this is the assertion that no integral bidegree- $(5, 35)$ curve $f(x, z) = 0$, with the boundary normal form and tangent constraints above, is the normalization of a profile-4 pair satisfying*

$$d \log r = z d \log(x - 1)$$

for some function r with the prescribed divisor. The assertion includes the lower-degree boundary strata obtained from this representative by imposing additional high-point coincidences compatible with the quotient table.

7. THE NO-HIGHPOINT BRANCH

If none of the six high points $P_0, P_1, P_\infty, Q_0, Q_1, Q_\infty$ agree, the low-degree quotient of Lemma 5.1 is unavailable. The residual approach below uses a bidegree- $(4, 4)$ ansatz. The first unproved part is that every no-highpoint solution is represented by this ansatz.

Let

$$\begin{aligned} P(X, R) = & R^4 X^4 - R^4 X^3 - R^3 X^4 + 2R^3 X - 2R^2 X \\ & + 2RX^3 - 2RX^2 + R + X - 1, \end{aligned}$$

and

$$L = 4XR + 4X + 4R + 2.$$

The line $L = 0$ is the graph

$$R = \frac{4X + 2}{X + 1},$$

which sends $X = 0, 1, \infty$ to $R = 2, 3, 4$, respectively. The boundary preserving residual family is

$$H_A = L^{31}P - X(X - 1)R(R - 1)A(X, R),$$

where

$$\deg_X A \leq 32, \quad \deg_R A \leq 32.$$

The factor $X(X - 1)R(R - 1)$ makes the six affine boundary restrictions agree with those of $L^{31}P$. The condition that the $(X, R) = (0, \infty)$ and $(\infty, 0)$ corners have multiplicity three forces the two corner coefficients

$$[R^{32}]A(0, R) = 2, \quad [X^{32}]A(X, 0) = 2.$$

Given the boundary profile, Riemann–Hurwitz says that a valid residual curve has no interior ramification for either projection: the three order-31 boundary branches already use the full ramification budget. Thus a residual candidate must satisfy, away from the boundary,

$$H_A = \frac{\partial H_A}{\partial R} = 0 \quad \text{has no solution}$$

for the X -projection, and similarly with X and R interchanged. In characteristic 5, writing $D = X(X - 1)R(R - 1)$ and $F = LP$, the ramification equations away from $F = 0$ may be written as

$$N_R = F \partial_R(DA) - DA \partial_R F = 0,$$

$$N_X = F \partial_X(DA) - DA \partial_X F = 0.$$

Several small subfamilies of this residual system have been eliminated: constant A , sparse low-parameter families, and the symmetric ordinary-corner candidates. These eliminations are not enough for the full coefficient space, which has 33^2 coefficients before boundary conditions, and they are not used as proof of the full no-highpoint branch below.

Open theorem 7.1 (C. No-highpoint exclusion). *No profile-4 two-map pair exists with all six high points*

$$P_0, P_1, P_\infty, Q_0, Q_1, Q_\infty$$

distinct.

A sufficient residual route to this theorem is the conjunction of the following two precise statements:

- (1) *every no-highpoint profile-4 pair gives a residual equation $H_A = 0$ of the form above;*
- (2) *no polynomial A in that residual family satisfies all boundary, integrality, genus, and no-interior-ramification conditions over $\overline{\mathbb{F}}_5$.*

8. THE ENTRY-ZERO BRANCH

By Lemma 5.2, it is enough to study the entry-zero representative

$$Q_0 = P_0.$$

By Corollary 5.4, after the entry-one branch is removed, this representative has no other high-point coincidence.

8.1. The quotient and its norms. Assume $Q_0 = P_0$ and no other high-point coincidence. Define

$$z = \frac{\Omega_0(r)}{\Omega_0(x)} = \frac{dr/(r-1)}{dx/(x-1)}.$$

Then

$$\operatorname{div}(z) = P_1 + P_\infty + C + F - Q_1 - Q_\infty - A - B,$$

so $\deg z = 6$. Since $\gcd(35, 6) = 1$, the field $k(x, z)$ is all of $k(C)$. Therefore C is the normalization of an integral bidegree $(6, 35)$ curve $f(x, z) = 0$ in $\mathbb{P}_x^1 \times \mathbb{P}_z^1$.

Let

$$\alpha = z(P_0), \quad c = x(Q_1), \quad d = x(Q_\infty),$$

and set

$$s = c + d, \quad p = cd, \quad N = (1 - c)(1 - d) = 1 - s + p.$$

The no-other-highpoint assumption and the fact that U is the common unramified boundary imply

$$c, d \in k \setminus \{0, 1\};$$

the points Q_1, Q_∞ are distinct, but the values c, d need not be distinct. Normalize f by

$$[x^6]f = z^{34}(z + 1), \quad [z^{35}]f = x^4(x - c)(x - d).$$

The boundary fibers have the form

$$\begin{aligned} f(x, 0) &= K(x - 1)^2, \\ f(0, z) &= K(1 - \eta z)^{31}, \quad \eta = \alpha^{-1}, \end{aligned}$$

and

$$f(1, z) = Nz^{32}(z - 1)^2(z + 1).$$

These formulas are just the pushed-forward divisors of $z, x, x - 1$. Explicitly,

$$\operatorname{Norm}_{C/k(x)}(z) = \frac{(x - 1)^2}{x^4(x - c)(x - d)},$$

$$\operatorname{Norm}_{C/k(z)}(x) = \frac{(z - \alpha)^{31}}{z^{34}(z + 1)},$$

and

$$\text{Norm}_{C/k(z)}(x-1) = \frac{(z-1)^2}{z^2}.$$

Lemma 8.1 (Norm-constant reduction). *In the entry-zero no-other-highpoint branch,*

$$\alpha^{31} = 1, \quad (1-c)(1-d) = 1, \quad cd = -1.$$

Consequently

$$c = d = 2.$$

Proof. Since f has odd degree 35 in z ,

$$\text{Norm}_{C/k(x)}(z) = -\frac{f(x, 0)}{x^4(x-c)(x-d)} = -\frac{K(x-1)^2}{x^4(x-c)(x-d)}.$$

Comparison with the exact norm gives $K = -1$.

Since f has degree 6 in x ,

$$\text{Norm}_{C/k(z)}(x) = \frac{f(0, z)}{z^{34}(z+1)} = \frac{K(1-\eta z)^{31}}{z^{34}(z+1)}.$$

As $1 - \eta z = -\eta(z - \alpha)$, comparison gives

$$K(-\eta)^{31} = 1.$$

With $K = -1$, this implies $\eta^{31} = 1$, hence $\alpha^{31} = 1$.

Similarly,

$$\text{Norm}_{C/k(z)}(x-1) = \frac{f(1, z)}{z^{34}(z+1)} = N \frac{(z-1)^2}{z^2},$$

so $N = 1$.

It remains to identify $p = cd$. Put $w = 1/z$ and

$$\bar{f}(x, w) = w^{35} f(x, 1/w).$$

At $(x, w) = (0, 0)$, the residues of $(dr/(r-1))/(dx/(x-1))$ give $z \sim -1/x$ on the atom $A = U_{0,1}$, while the three points of $B = U_{0,\infty}$ have $z \sim 1/x$. Hence the degree-four tangent cone is proportional to

$$(w+x)(w-x)^3.$$

The w^4 -coefficient of the degree-four part of $\bar{f}$ is $K(-\eta)^{31}$, and the x^4 -coefficient is p . In $(w+x)(w-x)^3$, these two coefficients have opposite signs. Therefore

$$p = -K(-\eta)^{31} = -1.$$

Since $N = 1 - s + p = 1$ and $p = -1$, we have $s = -1$. The numbers c, d are the roots of

$$T^2 - sT + p = T^2 + T - 1.$$

Over characteristic 5, this is $(T-2)^2$. Hence $c = d = 2$. □

Thus the entry-zero branch is reduced to the corrected double-fiber case

$$Q_0 = P_0, \quad x(Q_1) = x(Q_\infty) = 2, \quad Q_1 \neq Q_\infty.$$

8.2. The formal local tower. The double-fiber case is studied by expanding the normalized bidegree (6, 35) equation at the double fiber $x = 2$ and imposing the characteristic-5 differential equation attached to

$$z = \frac{dr/(r-1)}{dx/(x-1)}.$$

The expansion alternates repeated and simple layers. The layers through simple u^{25} , together with the repeated u^{25} two-minor step, have explicit linear certificates: at each layer the relevant residual equations either solve the next group of coefficients or cut to a smaller affine subspace by a displayed full-rank matrix. The first unresolved global local frontier is the tail-exhausted simple- u^{30} all-zero locus together with the base residual and repeated- $e30$ consistency.

One checked component of this frontier is the point $P129$. In local coordinates, the representative is

$$(2, 1, 2, 3, 4, 3, 2, 1, 1, 3, 0, 2, 0, 2, 4, 2, 1, 1, 2).$$

After the repeated- u^{25} pivot solve, the remaining free variables are

$$u7_3, u6_2, u6_3, u5_1, u5_2, u4_0, u4_1, u4_2, u4_3, \\ u3_0, u3_1, u3_2, u3_3, u2_2, u1_0, u1_1.$$

The repeated- $e30$ condition is the affine three-row system

$$(3, 2, 4) + \sum_{\nu=1}^{16} y_{\nu} c_{\nu} = 0,$$

where the columns c_{ν} , in the order above, are

variable	c_{ν}
$u7_3$	(1, 4, 1)
$u6_2$	(4, 3, 3)
$u6_3$	(0, 1, 1)
$u5_1$	(2, 2, 4)
$u5_2$	(0, 1, 3)
$u4_0$	(2, 3, 3)
$u4_1$	(2, 1, 2)
$u4_2$	(0, 1, 4)
$u4_3$	(3, 1, 1)
$u3_0$	(1, 4, 4)
$u3_1$	(1, 1, 1)
$u3_2$	(4, 3, 0)
$u3_3$	(1, 4, 3)
$u2_2$	(0, 3, 2)
$u1_0$	(1, 0, 0)
$u1_1$	(3, 3, 3).

This matrix has rank 3, leaving an affine 13-space. The later layers $e35, e40, e45$ have ranks

$$4, \quad 4, \quad 4$$

on successive solution spaces, leaving a final line. On that line the simple $e50$ residual is the nonzero constant 2. Therefore $P129$ does not extend to a formal local solution through $e50$. The same $e50$ ending also kills the checked nearby tangent points of $P129$ and the checked $\rho4$ tangent hit.

8.3. Basin 127. Another high-nullity component of the same frontier is a six-parameter affine family with coordinates $r_0, \dots, r_5$. The expected saturated quotient is the ideal

$$\begin{aligned} J = \langle & r_2 - 2, \ r_5 - 2, \ r_0 + 2r_1 - 2r_3 + 2r_4^2 + r_4 + 1, \\ & r_1^2 - r_3 - 2r_4 - 1, \ r_1r_3 + 2r_1 - 2r_3 - r_4^2 + 2r_4, \\ & r_1r_4 - r_1 + 2r_4^2 - 2, \ r_3^2 + r_3 - 2, \ r_3r_4 - r_3 - r_4 + 1, \\ & r_4^3 - 2r_4^2 - r_4 + 2 \rangle \subset \mathbb{F}_5[r_0, \dots, r_5]. \end{aligned}$$

Lemma 8.2 (The displayed basin-127 quotient). *The ideal J is radical of length 5. Its five points are*

$$\begin{aligned} (0, 0, 2, 1, 4, 2), \quad (0, 1, 2, 3, 1, 2), \quad (3, 4, 2, 1, 2, 2), \\ (4, 2, 2, 1, 1, 2), \quad (4, 4, 2, 3, 1, 2). \end{aligned}$$

Moreover, adding the base residual

$$B_{127} = r_3 + r_4^2.$$

cuts J to the single maximal ideal

$$(r_0 + 2, \ r_1 + 1, \ r_2 - 2, \ r_3 - 1, \ r_4 - 2, \ r_5 - 2),$$

which is the already killed P129 point.

Proof. The generators first give $r_2 = r_5 = 2$ and

$$r_4^3 - 2r_4^2 - r_4 + 2 = (r_4 - 1)(r_4 - 2)(r_4 + 1).$$

Also

$$r_3r_4 - r_3 - r_4 + 1 = (r_4 - 1)(r_3 - 1).$$

If $r_4 = 4 = -1$, then $r_3 = 1$. The equation

$$r_1r_4 - r_1 + 2r_4^2 - 2 = 0$$

gives $r_1 = 0$, and the linear equation for r_0 gives $r_0 = 0$. If $r_4 = 2$, then $r_3 = 1$, the same equation gives $r_1 = 4$, and then $r_0 = 3$.

It remains to set $r_4 = 1$. Then $r_3^2 + r_3 - 2 = 0$, so

$$r_3 \in \{1, 3\}.$$

For $r_3 = 1$, the equation

$$r_1r_3 + 2r_1 - 2r_3 - r_4^2 + 2r_4 = 0$$

gives $r_1 = 2$, and then $r_0 = 4$. For $r_3 = 3$, the equation

$$r_1^2 - r_3 - 2r_4 - 1 = 0$$

gives $r_1 \in \{1, 4\}$, and the linear equation gives respectively $r_0 = 0$ and $r_0 = 4$. These are exactly the five displayed points.

All factors in this triangular decomposition are squarefree over $\mathbb{F}_5$, so J is the intersection of the corresponding maximal ideals. Evaluating $B_{127} = r_3 + r_4^2$ on the five points leaves only $(3, 4, 2, 1, 2, 2)$, which is the maximal ideal displayed above. $\square$

Certificate target 8.3 (D1. Basin-127 containment). *Let I_{127} be the ideal generated by the basin-127 low-data equations in the six coordinates $r_0, \dots, r_5$, and let s_{127} be the product of the etale discriminant and the chart pivot. Then*

$$V(I_{127}) \cap D(s_{127}) \subseteq V(J),$$

or equivalently every generator of $\sqrt{J}$ vanishes on the localized low-data scheme. The stronger saturated equality

$$I_{127} : s_{127}^\infty = J$$

implies this containment.

Target 8.3 is the part presently being checked in Sage. It would close the basin-127 component by reducing its base-residual locus to the already killed $P129$ point.

Certificate target 8.4 (D2. Double-fiber tail coverage). *On each $\rho3$ and $\rho4$ pivot chart, form the localized ideal*

$$I_{\text{chart}} = \langle \Phi_0, \Phi_1, \Phi_2, S_{30}, E30_0, E30_1, E30_2 \rangle : s_{\text{chart}}^\infty,$$

where Φ_i are the shared repeated low-data equations, S_{30} is the simple- $u30$ Schur/base residual, $E30_i$ are the three affine repeated $e30$ equations after the repeated- $u25$ pivot solve, and s_{chart} is the product of the etale discriminant and all chart pivots. Then $V(I_{\text{chart}})$ is contained in the union of the $P129$, $\rho4$, and basin-127 strata, and after the corresponding $e35, e40, e45, e50$ layer equations are adjoined on each stratum, the remaining localized ideals are unit ideals.

Open theorem 8.5 (D. Double-fiber completion). *There is no profile-4 pair satisfying*

$$Q_0 = P_0, \quad x(Q_1) = x(Q_\infty) = 2, \quad Q_1 \neq Q_\infty,$$

with no other high-point coincidence. It follows from the two certificate targets above.

9. ASSEMBLY OF THE CURVE-LEVEL REDUCTION

Proposition 9.1 (Certificate targets imply Open Theorem D). *Assume Certificate Targets 8.3 and 8.4. Then Open Theorem 8.5 holds.*

Proof. By Lemma 8.1, every entry-zero no-other-highpoint solution lies in the corrected $c = d = 2$ double-fiber branch. The formal layer certificates through repeated u^{25} and simple u^{30} reduce this branch to the tail-exhausted simple- $u30$ all-zero locus with base residual and repeated- $e30$ consistency. The $P129$ and $\rho4$ local strata are killed at $e50$. Target 8.3 reduces the basin-127 base-residual part to $P129$, already killed. Target 8.4 says that every other point of the localized frontier is contained in one of these killed strata or becomes empty after the later layer equations. $\square$

Proposition 9.2 (Open Theorems B–D eliminate profile 4). *Assume Open Theorems 6.1, 7.1, and 8.5. Then no profile-4 two-map pair exists.*

Proof. Let (C, x, r) be a profile-4 pair. If there is no high-point coincidence, Open Theorem 7.1 removes it. If there is an entry-one high-point coincidence, Open Theorem 6.1 removes it. The high-point quotient table removes the three coincidences

$$Q_\infty = P_0, \quad Q_1 = P_1, \quad Q_0 = P_\infty.$$

Thus any remaining high-point coincidence is entry-zero. By simultaneous inversion it is represented by $Q_0 = P_0$. After entry-one cases have been removed, Corollary 5.4 says there is no other high-point coincidence. Lemma 8.1 reduces this branch to the corrected double-fiber case, and Open Theorem 8.5 removes it. Therefore no profile-4 pair remains. $\square$

Theorem 9.3 (Visibility from the four open theorems). *Assume Open Theorems 3.5, 6.1, 7.1, and 8.5. Then every finite etale self-correspondence of $S = \mathbb{P}_k^1(31, 31, 31)$ is visible. Equivalently, $\text{Comm}_{\text{alg}, k}(S) = \text{Aut}(S) = S_3$.*

Proof. Let $u, v : T \rightarrow S$ be a connected representable finite etale self-correspondence. If

$$\pi_0 \circ u \not\simeq \pi_0 \circ v,$$

then Open Theorem 3.5 produces a profile-4 two-map pair. This contradicts Proposition 9.2. Hence $\pi_0 \circ u \simeq \pi_0 \circ v$ for every connected finite etale self-correspondence of S , so every such correspondence is visible by Definition 2.3. $\square$

Corollary 9.4 (Conditional common-cover counterexample from the four open theorems). *If Open Theorems 3.5, 6.1, 7.1, and 8.5 are proved, then*

$$Y : y^{31} = x(x - 1)$$

has no connected common finite etale cover with any smooth proper connected genus-three curve over k .

Proof. Combine Theorem 9.3 with Proposition 2.5. □

10. OPEN THEOREMS

The reduction leaves the following four independent statements.

- (A) Open Theorem 3.5: every non-visible finite etale self-correspondence of S produces the profile-4 degree-35 pair.
- (B) Open Theorem 6.1: no profile-4 pair has an entry-one high-point coincidence.
- (C) Open Theorem 7.1: no profile-4 pair has all six high points distinct.
- (D) Open Theorem 8.5: the corrected entry-zero double-fiber branch is empty.

The Sage computation concerns item (D), more specifically Certificate Target 8.3. If it proves

$$V(I_{127}) \cap D(s_{127}) \subseteq V(J)$$

or the stronger saturation equality

$$I_{127} : s_{127}^\infty = J,$$

then the basin-127 component of the double-fiber frontier is closed: the base residual cuts it to P_{129} , already killed at e_{50} . That still leaves Certificate Target 8.4, the chart-wise coverage of the rest of the tail-exhausted simple- u_{30} base-zero/ e_{30} locus.

Thus the computation is one algebraic containment inside Open Theorem 8.5. Together with Open Theorems 3.5, 6.1, and 7.1, Open Theorem 8.5 implies the visibility theorem by Theorem 9.3.